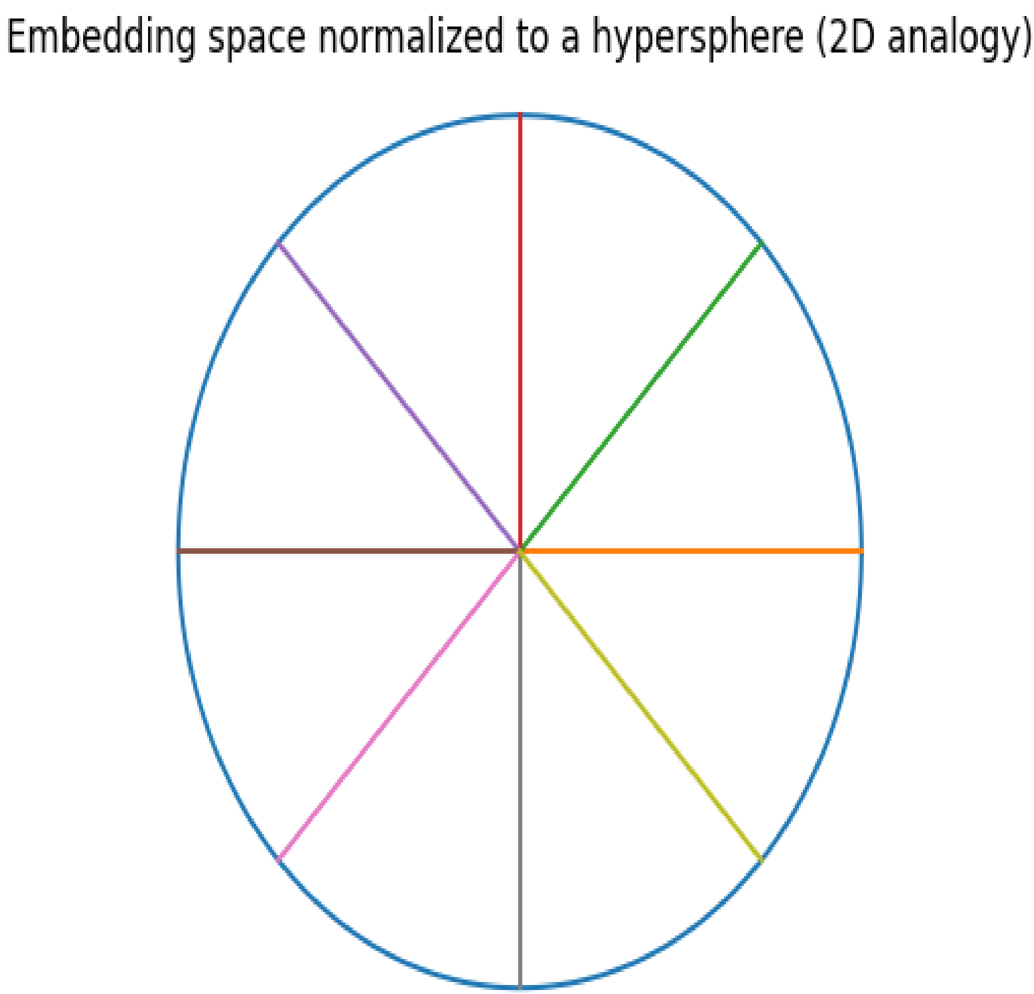
Angular Manifold Routing — Visual Explainer

This document visually explains the Angular Manifold Routing architecture. The goal is to show how geometric routing leads to sparse computation and improved scaling.

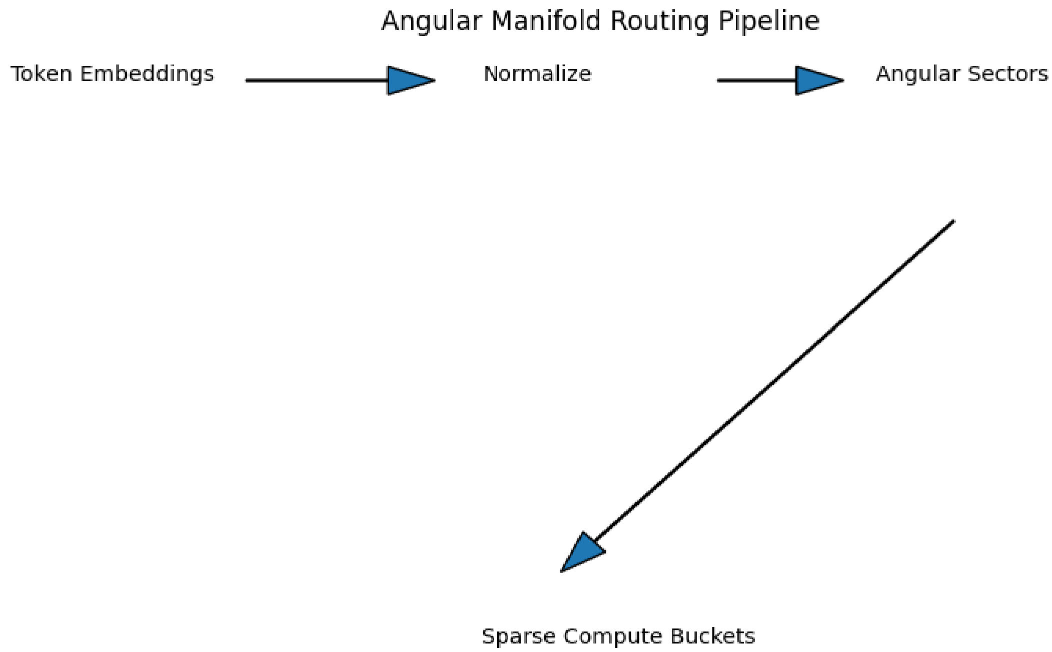
1. Embeddings on a Hypersphere

Embeddings are normalized so that all tokens lie on the surface of a hypersphere. Only the direction of the vector matters, not its magnitude. Routing decisions are made using angular sectors.



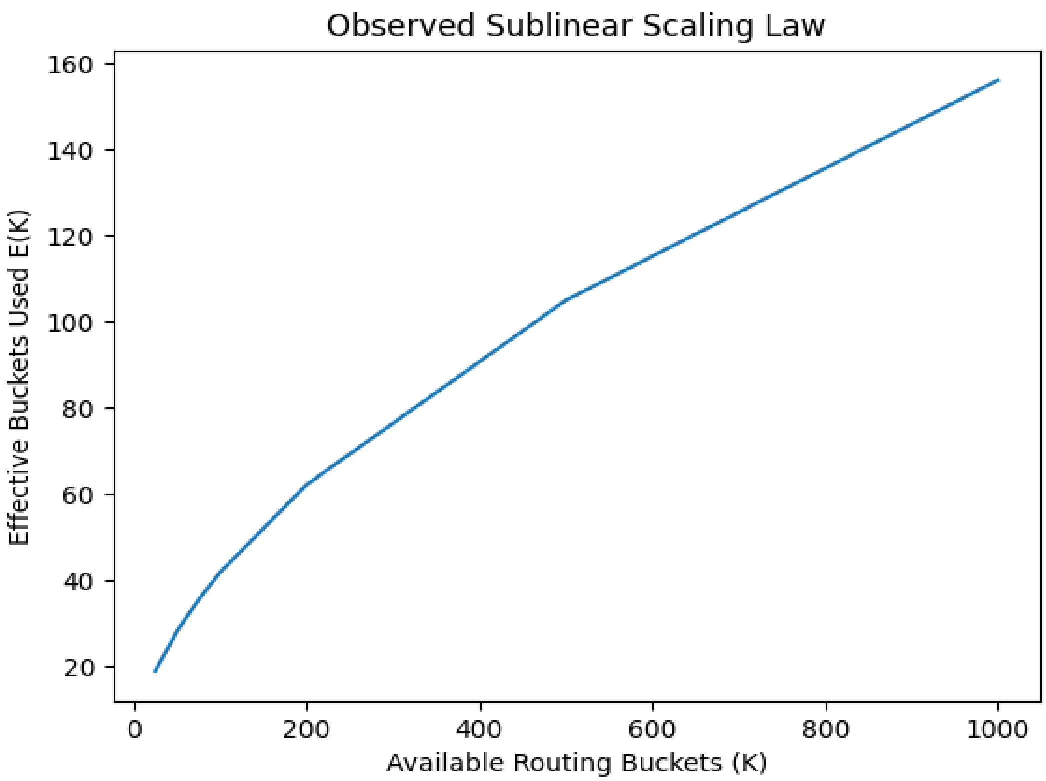
2. Angular Routing Pipeline

Tokens are embedded, normalized, and routed according to their angular position. Each angular sector represents a routing bucket. Only sectors that receive tokens activate compute modules.



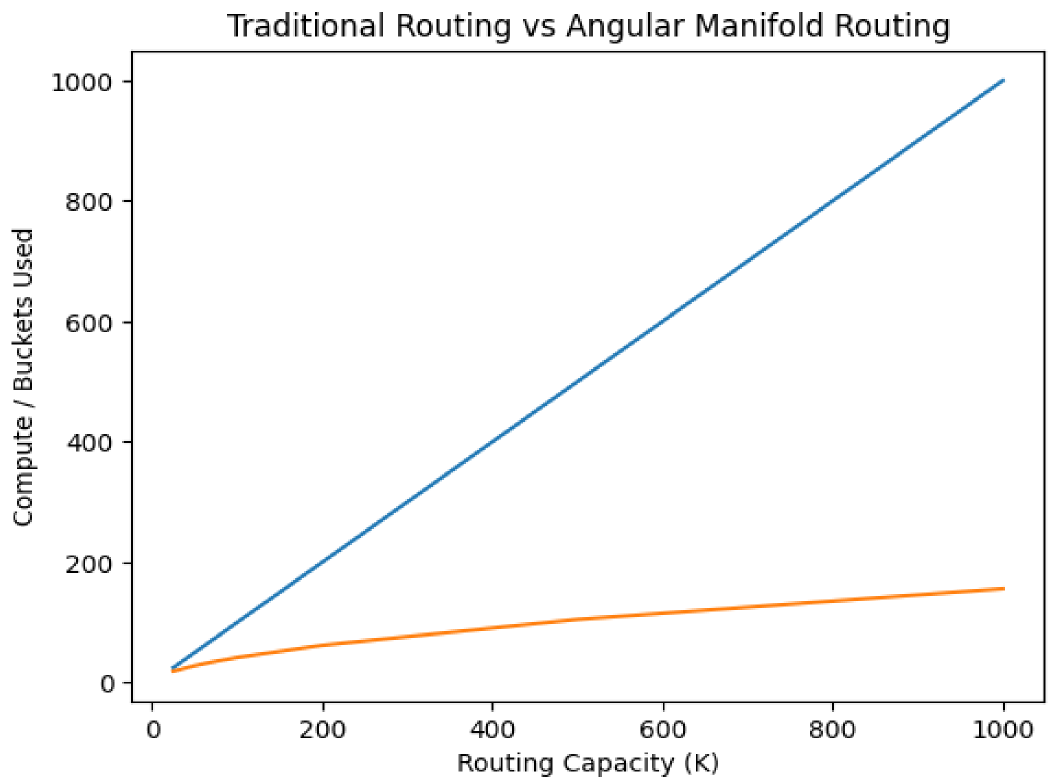
3. Sublinear Scaling Law

Experiments show that the number of active routing buckets grows sublinearly. The empirical law observed in experiments is: $E(K) \approx c \cdot K^{0.572}$



4. Comparison With Traditional Routing

Traditional routing scales linearly with routing capacity. Angular Manifold Routing grows more slowly, meaning fewer buckets are used even as routing capacity increases.



This geometric routing effect allows models to concentrate computation into fewer active buckets, improving hardware locality and reducing effective compute cost.